Research Paper

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Project Title:

CUSTOMER ENGAGEMENT & PRODUCT UTILIZATION ANALYTICS FOR RETENTION STRATEGY

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ABSTRACT

Customer retention is an important concern for banking institutions because losing existing customers can affect customer relationships and long-term business performance. Traditional customer analysis often emphasizes demographic and financial characteristics, but these characteristics alone may not fully explain why customers leave a bank. Customer engagement, product utilization, and the strength of the banking relationship may also provide useful insights into retention patterns.

This project, titled *Customer Engagement & Product Utilization Analytics for Retention Strategy*, examines the relationship between customer activity, banking product usage, account balance, credit card ownership, and customer churn using a European banking dataset. The dataset contains 10,000 customer records and 14 original source columns. Data validation, exploratory data analysis, customer segmentation, and feature engineering were performed using Python. Six additional analytical columns were created to support engagement classification, balance segmentation, product utilization analysis, and relationship-strength assessment.

The analysis identified an overall churn rate of 20.37%. Inactive customers had a churn rate of 26.85%, compared with 14.27% among active customers. Customers holding two banking products had a churn rate of 7.58%, while the rates for customers holding three and four products were considerably higher. High-balance customers also showed a higher churn rate than customers in the lower-balance group. In addition, retained customers had a higher average Relationship Strength Index than churned customers.

An interactive Streamlit dashboard was developed to present the results through key performance indicators, visualizations, customer filters, and a high-value disengaged customer analysis module. The findings demonstrate the value of examining engagement and product utilization together when studying banking customer retention. However, the results are descriptive and do not establish causal relationships. Further statistical testing

and validation would be necessary before applying the findings to operational retention decisions.

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1. INTRODUCTION

Customer retention is an important component of business strategy in the banking industry. Banks establish long-term relationships with customers through savings accounts, credit cards, loans, and other financial services. Maintaining these relationships can support customer satisfaction, business continuity, and sustainable growth.

However, customers may discontinue their banking relationships for various reasons, including low engagement, dissatisfaction with services, changing financial needs, limited product utilization, or the availability of alternative banking services. Understanding customer churn is therefore important for developing effective retention strategies.

Customer churn refers to the situation in which a customer ends their relationship with a business. In the banking context, churn occurs when a customer leaves the bank, as represented by the churn indicator available in the dataset used for this study.

Customer demographics and financial characteristics provide useful information about customer profiles, but they may not be sufficient to understand retention patterns. For example, a customer with a high account balance may still be inactive, while a customer who regularly engages with banking services may demonstrate a different retention pattern.

Similarly, the number of banking products held by a customer may indicate the depth of the banking relationship. However, a greater number of products does not automatically guarantee loyalty. The relationship between product utilization and retention must be examined using actual customer data.

This project focuses on understanding customer retention from a behavioural and relationship-strength perspective. It analyses customer engagement, product utilization, account balance, credit card ownership, and churn status to identify patterns that may help banks understand customer retention more effectively.

The project also uses data visualization and an interactive Streamlit dashboard to communicate findings in an accessible format.

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2. BACKGROUND AND CONTEXT

Banking institutions collect customer information related to financial activity, product ownership, demographics, and account status. These data can be analysed to understand customer behaviour and identify patterns associated with churn.

Despite the availability of customer data, identifying customers who may leave a bank remains a business challenge. Retention strategies based only on broad customer categories may fail to account for differences in engagement and product utilization.

For example, two customers may have similar account balances but different levels of engagement with the bank. One may actively use banking services, while the other may

have limited interaction. These differences may be relevant when examining retention patterns.

Product ownership is another important consideration. Customers with multiple banking products may have different relationships with the bank from customers who use only one product. Nevertheless, the relationship between product count and retention may not be straightforward.

Analysing these factors together can provide a more detailed understanding of customer behaviour. Such analysis can support the development of targeted customer engagement initiatives, product optimization strategies, and retention programs.

The present study therefore examines customer retention through the combined perspectives of engagement, product utilization, and financial characteristics.

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3. PROBLEM STATEMENT

Banking institutions require reliable information to understand customer churn and develop effective retention strategies. However, customer retention decisions may be limited when analysis focuses primarily on demographic information or financial characteristics without adequately considering customer engagement and product utilization.

A high account balance does not necessarily indicate strong customer engagement, and ownership of multiple banking products does not always correspond to a lower churn rate. Without examining these relationships, banks may find it difficult to identify customer groups that require additional attention.

The problem addressed by this project is the need to analyse customer engagement, banking product utilization, and financial characteristics together to understand their relationship with customer retention.

The study uses exploratory data analysis and customer segmentation to identify patterns in the available dataset. It also develops an interactive dashboard to present these findings and support data-informed discussions about retention strategies.

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4. OBJECTIVES OF THE STUDY

4.1 Primary Objectives

The primary objectives of the project are:

1. To examine the relationship between customer engagement and churn.
2. To analyse churn rates across different banking product counts.
3. To compare the retention patterns of customers with different account balances.
4. To identify high-balance customers who are classified as inactive.
5. To develop a Relationship Strength Index combining engagement and product utilization.

6. To create an interactive Streamlit dashboard for visualizing customer retention patterns.
7.

4.2 Secondary Objectives

The secondary objectives are:

1. To support engagement-oriented customer retention strategies.
2. To explore how product utilization relates to customer retention.
3. To identify customer segments that may warrant further investigation.
4. To present analytical findings through accessible visualizations.
5. To provide recommendations based on observed customer behaviour.
6. To demonstrate the practical application of Python in business and banking analytics.
7.

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5. SCOPE OF THE STUDY

The study focuses on customer retention patterns within the available European banking dataset.

The analysis covers customer engagement status, number of banking products, account balance, credit card ownership, and customer churn status. It also includes customer segmentation and a custom Relationship Strength Index.

Python was used for data preparation, analysis, and visualization. A Streamlit application was developed to present selected results interactively.

The scope of the project is primarily descriptive and exploratory. It identifies relationships and patterns within the dataset but does not establish that a particular customer characteristic causes churn.

The findings are intended to support further investigation and inform the design of potential retention strategies. They should not be interpreted as validated predictions of future customer behaviour.

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6. DATASET DESCRIPTION

6.1 Overview of the Dataset

The project uses a European banking customer dataset containing 10,000 records. Each record represents a customer and includes information about customer characteristics, banking relationships, financial details, and churn status.

The original dataset contains 14 source columns. During the analytical process, six additional columns were created to support customer segmentation and relationship-strength analysis. Consequently, the working Data Frame contains 20 columns.

The original dataset includes the following variables.

Variable	Description
Year	Year-related field provided in the dataset
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Customer credit score
Geography	Customer's recorded geography
Gender	Customer's recorded gender
Age	Customer age
Tenure	Length of the customer's relationship with the bank, as recorded in the dataset
Balance	Customer account balance
NumOfProducts	Number of banking products held
HasCrCard	Indicates whether the customer has a credit card
IsActiveMember	Indicates whether the customer is classified as active
EstimatedSalary	Estimated customer salary
Exited	Indicates whether the customer has churned

The Exited variable is the primary outcome variable used in the churn analysis. A value of 1 indicates a churned customer, while a value of 0 indicates a retained customer.

The IsActiveMember variable is used to distinguish active and inactive customers

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6.2 Data Validation

Initial data validation produced the following results.

Validation measure	Result
Total customer records	10,000
Original source columns	14
Columns after feature engineering	20
Missing values	0

Validation measure	Result
Duplicate rows	0
Unique customer IDs	10,000
Retained customers	7,963
Churned customers	2,037
Overall churn rate	20.37%

No missing values or duplicate rows were identified in the initial validation. All 10,000 customer records had unique customer IDs.

The absence of missing values and duplicate records supported the subsequent exploratory analysis. However, these checks alone do not establish that every field is accurate or that the dataset represents all banking customers.

6.3 Geographical Distribution

The geographical distribution of the customer records was as follows.

Geography	Number of customers
France	5,014
Germany	2,509
Spain	2,477
Total	**10,000**

France accounted for the largest number of customer records, followed by Germany and Spain.

This distribution provides context for the dataset. Although geography is available as a variable, the main emphasis of this project is customer engagement and product utilization rather than demographic segmentation.

6.4 Distribution by Number of Products

The distribution of customers according to the number of banking products held was:

Number of products	Number of customers
1	5,084
2	4,590

Number of products	Number of customers
3	266
4	60
Total	**10,000**

Most customers held either one or two banking products. Comparatively few customers held three or four products.

This distribution is important when interpreting churn rates. Results for product groups containing relatively few customers may be less stable and should be examined carefully.

6.5 Distribution by Engagement Status

Customer engagement was represented by the IsActiveMember variable.

Engagement status	Number of customers
Active	5,151
Inactive	4,849
Total	**10,000**

The dataset contains slightly more active customers than inactive customers.

This variable provides the basis for comparing churn rates across engagement groups.

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7. METHODOLOGY

7.1 Research Approach

The project follows a quantitative, descriptive, and exploratory approach. Numerical customer data were analysed to identify patterns associated with customer engagement, product utilization, account balance, and churn.

The analysis focuses on describing observed relationships rather than establishing causality.

7.2 Data Loading and Preparation

The dataset was loaded into a Python environment using Pandas.

Initial preparation involved examining the dataset's dimensions, column names, missing values, duplicate records, and customer identifiers.

The Exited variable was used to distinguish retained customers from churned customers. The IsActiveMember variable was used to classify customers as active or inactive.

Additional analytical columns were created to support customer segmentation and the calculation of engagement, product usage, and relationship-strength scores.

7.3 Exploratory Data Analysis

Exploratory data analysis was performed to understand the distribution of customers and identify differences between customer groups.

The analysis included:

- Overall customer churn and retention.
- Customer distribution by engagement status.
- Churn rate by number of banking products.
- Account balance and engagement comparisons.
- Customer segmentation by balance and engagement.
- Credit card ownership and retention.
- Relationship Strength Index comparisons.

Tables and visualizations were used to present the results in an understandable format.

7.4 Engagement Analysis

Customers were classified as active or inactive based on the IsActiveMember variable.

The churn rate for each group was calculated using the following formula:

$$\text{Churn Rate} = \frac{Number\ of\ churned\ customers}{Total\ customers\ in\ the\ group} * 100$$

The churn rates of active and inactive customers were then compared to examine the relationship between engagement status and customer exit.

7.5 Product Utilization Analysis

Product utilization was examined using the NumOfProducts variable.

Customers were grouped according to whether they held one, two, three, or four banking products. The number of customers and churn rate were calculated for each group.

This analysis was used to examine whether retention patterns differed according to the number of products held.

The interpretation also considered group size because the numbers of customers holding three or four products were substantially smaller than the numbers holding one or two products.

7.6 Financial Commitment and Engagement Analysis

Account balance and engagement status were examined together to identify customers who had relatively high balances but were classified as inactive.

The high-balance threshold was defined using the 75th percentile of account balance. Customers at or above this threshold were classified as high-balance customers, while the remaining customers were classified as lower-balance customers.

This classification created a group of 2,500 high-balance customers and a group of 7,500 lower-balance customers.

The analysis compared churn rates and inactivity levels between these groups.

The purpose was to investigate whether a relatively high account balance was associated with different retention patterns and whether inactive high-balance customers could be identified for further investigation.

7.7 Relationship Strength Index

A custom Relationship Strength Index was developed to combine engagement and product utilization into a single analytical measure.

The index used two component scores:

- Engagement Score
- Product Usage Score

The Engagement Score was calculated by assigning a score of 100 to active customers and 0 to inactive customers.

The Product Usage Score was calculated by dividing the number of products held by a customer by the maximum number of products observed in the dataset and multiplying the result by 100.

The Relationship Strength Index was calculated as follows:

$$RSI = 0.5(\text{Engagement Score}) + 0.5(\text{Product Usage Score})$$

The two component scores were given equal weights.

The resulting index was compared between retained and churned customers to investigate whether the combined measure differed across customer groups.

The index is a project-specific exploratory measure. It has not been independently validated as a measure of customer loyalty or as a predictive model.

7.8 Key Performance Indicators

Five indicators were considered in the project.

KPI	Purpose
Engagement Retention Ratio	Compares retention outcomes for active and inactive customers
Product Depth Index	Examines product utilization among customers, including retained customers
High-Balance Disengagement Rate	Measures the proportion of high-balance customers classified as inactive
Credit Card Stickiness Score	Examines the retention rate among credit card holders
Relationship Strength Index	Combines engagement and product usage into a custom analytical score

These indicators provide different perspectives on customer retention.

They are intended to support descriptive analysis and comparison rather than to serve as validated standalone predictors.

7.9 Streamlit Dashboard Development

An interactive Streamlit dashboard was developed to present the analysis in a user-friendly format.

The dashboard includes key customer metrics, engagement and churn comparisons, product utilization analysis, high-value disengaged customer analysis, and Relationship Strength Index visualizations.

Users can apply filters related to geography, engagement status, number of products, account balance, and estimated salary.

The dashboard also presents customer records for further inspection and provides a download option for the high-value disengaged customer table.

The purpose of the dashboard is to make the results easier to explore and communicate without requiring users to execute Python code.

8. EXPLORATORY DATA ANALYSIS

8.1 Overall Customer Churn

The dataset contains 10,000 customers, of whom 7,963 were retained and 2,037 churned.

The overall churn rate was calculated as follows:

$$\text{Overall Churn Rate} = \frac{2{,}037}{10{,}000} * 100$$

$$= 20.37\%$$

The retention rate was:

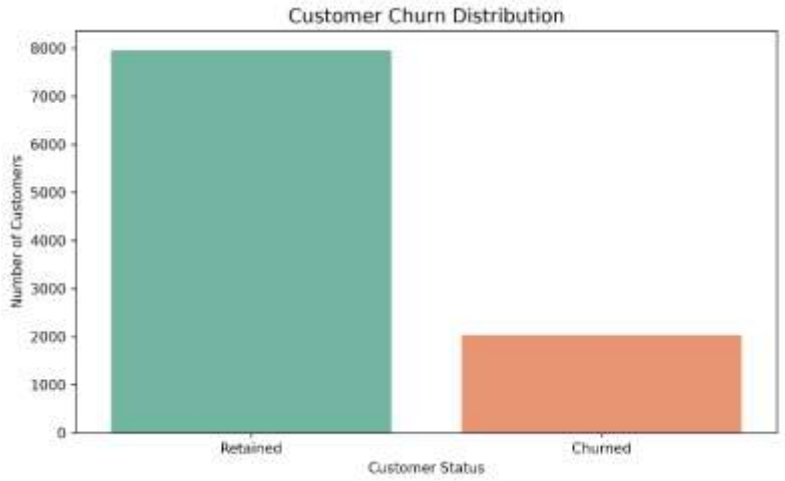
$$\text{Retention Rate} = \frac{7,953}{10,000} * 100$$

$$= 79.63\%$$

Therefore, approximately one-fifth of the customers in the dataset had churned, while the remaining customers were classified as retained.

This overall result provides a baseline for examining differences among customer groups.

Figure 1: Overall Customer Churn Distribution



8.2 Customer Engagement Distribution

The engagement analysis identified 5,151 active customers and 4,849 inactive customers.

Active customers accounted for 51.51% of the dataset, while inactive customers accounted for 48.49%.

The relatively balanced distribution of active and inactive customers makes it possible to compare the churn outcomes of these two groups.

However, engagement status alone does not explain every difference in customer behaviour. Other characteristics, such as product utilization and account balance, may also be relevant.

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9. CUSTOMER ENGAGEMENT AND CHURN ANALYSIS

Customer engagement was examined by comparing the churn rates of active and inactive customers.

Engagement status	Total customers	Churned customers	Churn rate
Inactive	4,849	1,302	26.85%
Active	5,151	735	14.27%

The results show that inactive customers had a churn rate of 26.85%, compared with 14.27% among active customers.

The difference between the two churn rates was:

$$26.85\% - 14.27\% = 12.58$$

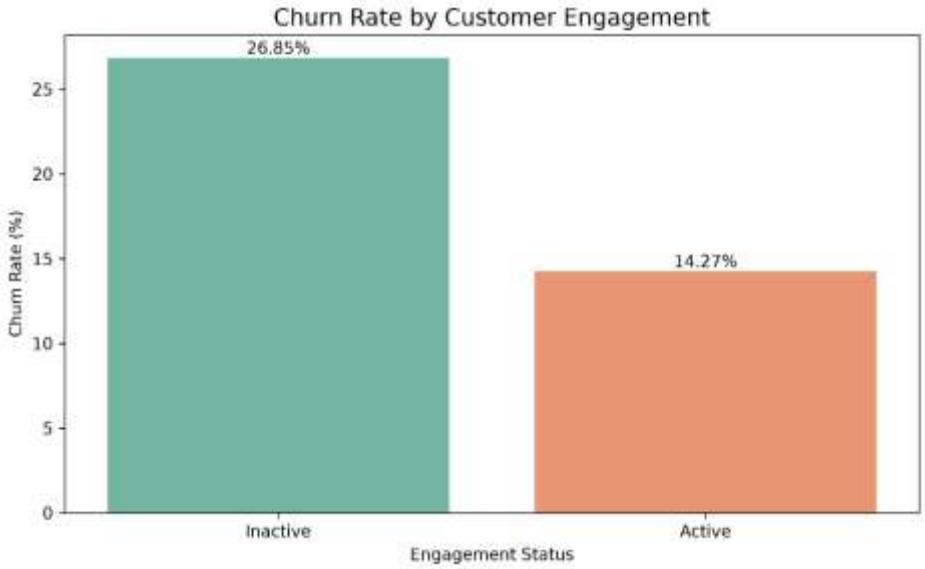
Therefore, the observed churn rate among inactive customers was 12.58 percentage points higher than that among active customers.

This finding indicates an association between inactivity and higher customer churn in the dataset.

The result suggests that engagement status may be a useful variable for identifying customer groups that warrant additional attention. For example, a bank could investigate whether inactive customers require improved communication, service reviews, or engagement initiatives.

However, the analysis does not establish that inactivity directly causes customers to leave. Other customer characteristics may also contribute to the observed difference.

Figure 2: Churn Rate by Customer Engagement



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10. PRODUCT UTILIZATION AND RETENTION ANALYSIS

Product utilization was examined by comparing churn rates among customers holding different numbers of banking products.

Number of products	Total customers	Churned customers	Churn rate
1	5,084	1,409	27.71%
2	4,590	348	7.58%

Number of products	Total customers	Churned customers	Churn rate
3	266	220	82.71%
4	60	60	100.00%

The results reveal substantial differences in churn across the product groups.

Customers holding two products had the lowest observed churn rate, at 7.58%. Customers holding one product had a churn rate of 27.71%.

In contrast, the churn rates among customers holding three and four products were 82.71% and 100%, respectively.

These results indicate that product count and customer retention do not have a simple, consistently decreasing relationship in this dataset.

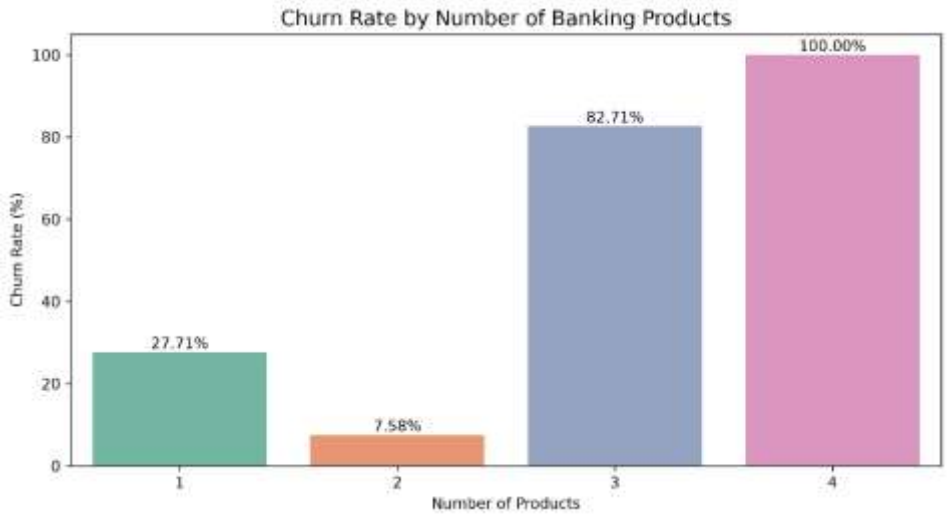
The high churn rates among customers holding three or four products require careful interpretation. These groups contained only 266 and 60 customers, respectively. The 100% churn rate in the four-product group reflects the observed outcomes of just 60 customers and should not automatically be generalized to other customer populations.

The findings also show why product count should not be treated as a direct measure of loyalty. The number of products held does not necessarily indicate how frequently or effectively customers use those products.

Further investigation could examine the characteristics of customers in each product group, their engagement status, and whether other variables are associated with their retention outcomes.

Business implication: Banking institutions should evaluate product utilization and customer engagement together when designing product bundling and retention strategies.

Figure 3: Churn Rate by Number of Banking Products



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11. HIGH-BALANCE DISENGAGEMENT ANALYSIS

The project examines whether customers with relatively high account balances are necessarily more engaged with the bank.

Customers were classified into two groups based on the 75th percentile of account balance.

Customer group	Total customers	Inactive customers	Churned customers	Churn rate
Lower Balance	7,500	3,602	1,445	19.27%
High Balance	2,500	1,247	592	23.68%

The high-balance group had a churn rate of 23.68%, compared with 19.27% for the lower-balance group.

The difference between these rates was:

$$23.68\% - 19.27\% = 4.41$$

Thus, the observed churn rate among high-balance customers was 4.41 percentage points higher.

The analysis also identified 1,247 inactive customers in the high-balance group.

The high-balance disengagement rate was calculated as follows:

$$\text{High Balance Disengagement Rate} = \frac{Inactive\ high\ balance\ customers}{Total\ high\ balance\ customers} * 100$$

$$= \frac{1{,}247}{2{,}500} * 100$$

High Balance Disengagement Rate = 49.88%

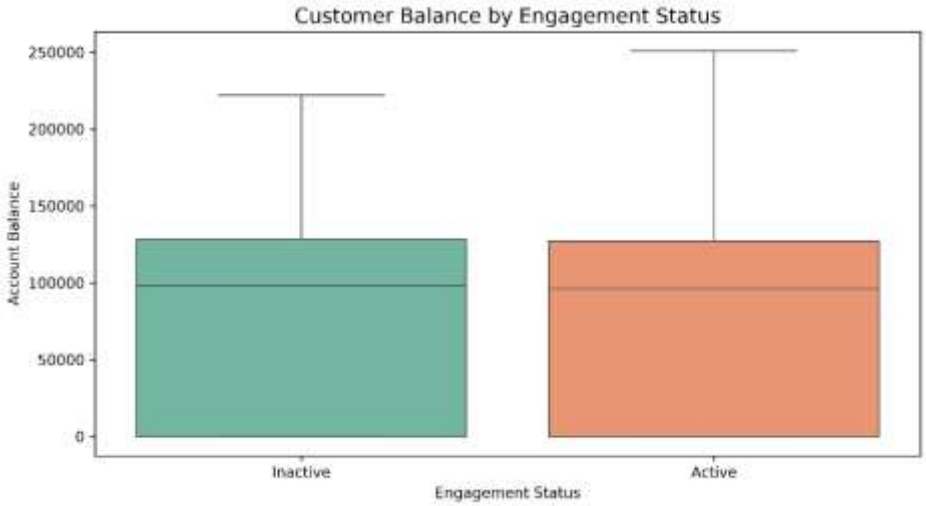
Therefore, 49.88% of customers in the high-balance group were classified as inactive.

These findings suggest that account balance alone may not provide a complete picture of the strength of a customer's relationship with a bank. A customer may maintain a high balance while showing limited engagement.

The identification of high-balance inactive customers may help banks determine which customer groups deserve further investigation. However, the classification does not establish that every inactive high-balance customer is at risk of leaving.

Business implication: Banks could combine account balance, engagement status, and churn-related indicators when reviewing customer retention opportunities.

Figure 4: Account Balance and Customer Engagement



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12. RELATIONSHIP STRENGTH INDEX ANALYSIS

12.1 Purpose of the Index

The Relationship Strength Index was developed to examine customer retention using a combined measure of engagement and product utilization.

The index was designed for this project to explore whether retained and churned customers differed in their average engagement-product scores.

The formula used was:

$$\text{RSI} = 0.5(\text{Engagement Score}) + 0.5(\text{Product Usage Score})$$

The index gives equal weight to engagement and product usage. It is an exploratory measure developed for this project rather than an established industry-standard metric.

12.2 Results of the RSI Analysis

The RSI results were as follows.

Customer status	Number of customers	Mean RSI	Median RSI
Retained	7,963	47.03	62.50
Churned	2,037	36.48	25.00

Retained customers had an average RSI of 47.03, while churned customers had an average RSI of 36.48.

The difference between the mean scores was:

$$47.03 - 36.48 = 10.55$$

Therefore, the mean RSI among retained customers was 10.55 points higher than the mean RSI among churned customers.

The median RSI was also higher among retained customers, at 62.50 compared with 25.00 among churned customers.

These results suggest that the combination of engagement and product usage distinguishes the two customer groups within the observed dataset.

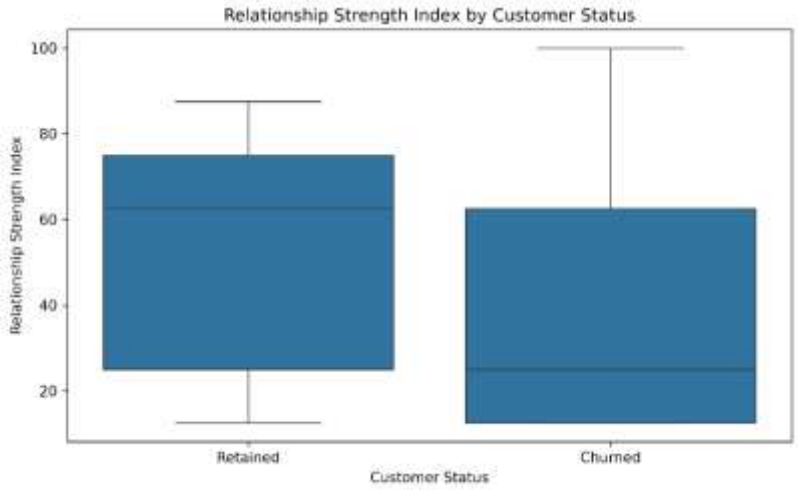
Nevertheless, the index should be interpreted carefully. Since engagement contributes substantially to the score and the weights were selected for this project, the difference in RSI may partly reflect the existing relationship between engagement status and churn.

The analysis does not independently validate the index as a customer loyalty measure or prove that increasing a customer's score would reduce churn.

Further testing would be necessary to assess the index's usefulness for operational decision-making.

Business implication: A combined engagement-product score may be useful as an exploratory segmentation measure, provided that it is validated before being used to guide retention decisions.

Figure 5: Relationship Strength Index by Customer Status



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13. CREDIT CARD OWNERSHIP AND RETENTION

Credit card ownership was examined to investigate whether retention rates differed between customers with and without a credit card.

Credit card status	Total customers	Retained customers	Retention rate
No Credit Card	2,945	2,332	79.19%
Has Credit Card	7,055	5,631	79.82%

The retention rate among customers with a credit card was 79.82%, compared with 79.19% among customers without a credit card.

The difference was:

$$79.82\% - 79.19\% = 0.63$$

Therefore, the retention rate was only 0.63 percentage points higher among customers with a credit card.

The observed difference is relatively small. Based on these descriptive results, credit card ownership alone does not appear to distinguish retention outcomes as clearly as customer engagement does.

However, the analysis does not establish whether the difference is statistically significant. Additional testing would be required to determine whether the observed difference is likely to reflect a meaningful association in the broader customer population.

Business implication: Credit card ownership should be evaluated alongside other customer characteristics rather than used as a standalone indicator of retention strength.

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14. DEVELOPMENT OF THE STREAMLIT DASHBOARD

14.1 Dashboard Overview

An interactive Streamlit dashboard was developed to present the findings of the project in a user-friendly format.

The dashboard connects the analytical results with a visual interface that allows users to examine customer groups, compare retention patterns, and explore customer records.

The application is designed to make the analysis accessible to users who may not have experience with Python programming.

14.2 Key Performance Indicators

The dashboard presents summary indicators, including:

- Total number of customers.
- Overall churn rate.
- Overall retention rate.
- Number of active customers.

These indicators provide a quick overview of the customer population and its retention outcomes.

14.3 Engagement and Churn Module

The engagement module compares churn outcomes for active and inactive customers.

It presents the differences in churn rates between the two groups and allows users to explore the relationship between engagement status and customer retention.

This module highlights the observed association between inactivity and higher churn.

14.4 Product Utilization Module

The product utilization module examines churn across different product counts.

Users can compare customer volumes and churn rates for customers holding one, two, three, or four products.

This module helps demonstrate that product count does not have a simple linear relationship with churn in the analysed dataset.

14.5 High-Value Disengaged Customer Module

The high-value disengaged customer module identifies customers with relatively high balances who are classified as inactive.

It presents information about the size of the group, its churn outcomes, and the corresponding customer records.

The dashboard also provides a facility to download the displayed customer table for further examination.

The module is intended to support exploratory customer segmentation rather than automatically classify every selected customer as certain to churn.

14.6 Relationship Strength Analysis Module

The Relationship Strength Index module compares the index across customer statuses.

The visualization allows users to examine the differences in average relationship-strength scores between retained and churned customers.

The module helps communicate the potential usefulness of combined engagement and product-usage indicators while recognizing that the index requires further validation.

14.7 Interactive Filters

The dashboard provides filters for selected customer characteristics, including:

- Geography.
- Engagement status.
- Number of products.
- Account balance.
- Estimated salary.

These filters allow users to explore selected customer groups and compare their characteristics.

The resulting views support interactive examination of the dataset rather than requiring users to modify the original analysis code.

14.8 Dashboard Contribution

The Streamlit dashboard converts the analysis into an interactive reporting tool.

Its main contribution is the presentation of customer retention indicators in a format that supports exploration, comparison, and communication.

The dashboard does not currently represent a validated machine-learning churn prediction system. Its purpose is to display descriptive analytics and exploratory relationship-strength measures.

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15. KEY FINDINGS

The analysis produced the following major findings.

15.1 Overall Churn

The dataset contains 2,037 churned customers out of 10,000 records, resulting in an overall churn rate of 20.37%.

15.2 Customer Engagement

Inactive customers had a churn rate of 26.85%, compared with 14.27% among active customers. The 12.58-percentage-point difference indicates a notable association between engagement status and churn.

15.3 Product Utilization

Customers holding two products had the lowest observed churn rate, at 7.58%. Customers holding three and four products had much higher churn rates, although the sizes of these groups were comparatively small.

15.4 High-Balance Customers

High-balance customers had a churn rate of 23.68%, compared with 19.27% among lower-balance customers. In addition, 49.88% of high-balance customers were classified as inactive.

15.5 Relationship Strength Index

Retained customers had a mean RSI of 47.03, compared with 36.48 among churned customers. The result indicates that the custom index differed across the two customer groups.

15.6 Credit Card Ownership

The retention rate was 79.82% among customers with credit cards and 79.19% among those without credit cards. The difference was relatively small.

15.7 Value of Interactive Analytics

The Streamlit dashboard provides an accessible interface for examining customer retention patterns, filtering customer groups, and exploring analytical results.

Taken together, these findings indicate that customer engagement and product utilization are useful dimensions for further investigation into banking retention.

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16. RECOMMENDATIONS

Based on the observed results, the following recommendations are proposed for banking institutions.

16.1 Monitor Customer Engagement

Banks could monitor engagement indicators to identify customers who have become inactive.

Because inactive customers in this dataset had a higher churn rate, engagement monitoring may help identify groups that deserve further investigation.

Any retention initiative should be evaluated to determine whether it improves customer experience and retention outcomes.

16.2 Investigate Inactive High-Balance Customers

Banks could examine customers who have relatively high balances but are classified as inactive.

Such customers may have different needs or service preferences from actively engaged customers. Further investigation could determine whether communication, service support, or other engagement initiatives are appropriate.

High balance and inactivity should be treated as indicators for review rather than proof that a customer intends to leave.

16.3 Evaluate Product Utilization Carefully

The observed relationship between product count and churn suggests that banks should avoid assuming that more products always mean stronger retention.

The particularly high churn rates among customers with three or four products warrant further investigation. Their customer characteristics, engagement levels, and product usage patterns could be examined in more detail.

Product bundling strategies should be evaluated based on customer needs and actual utilization rather than product count alone.

16.4 Combine Multiple Customer Indicators

Banks could explore combining engagement status, product utilization, account balance, and other relevant indicators when reviewing retention patterns.

A multi-factor approach may provide a more detailed picture than examining one characteristic in isolation.

However, any scoring system should be tested and validated before being used to make customer-level decisions.

16.5 Evaluate Credit Card Ownership in Context

The observed retention difference between customers with and without credit cards was small.

Therefore, credit card ownership should not be assumed to be a strong standalone retention indicator. Its usefulness should be evaluated alongside engagement and other customer characteristics.

16.6 Use Dashboards for Monitoring and Communication

Interactive dashboards can help communicate analytical results to business teams and other stakeholders.

Banks could use similar dashboards to monitor customer groups, review changes in engagement indicators, and evaluate retention initiatives.

Any operational dashboard should use consistent definitions, validated calculations, appropriate access controls, and regularly reviewed data.

16.7 Validate Indicators Before Operational Use

The Relationship Strength Index was developed as an exploratory measure for this project.

Before it is used to guide retention decisions, its relationship with customer outcomes should be evaluated using additional testing. Alternative scoring weights and additional customer characteristics could also be considered.

This would help determine whether the index provides useful information beyond the individual variables from which it is constructed.

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17. LIMITATIONS OF THE STUDY

Although the project provides useful descriptive insights, several limitations should be considered.

17.1 Dataset Limitations

The analysis is based on a single banking dataset containing 10,000 customer records. The results may not represent customers from other banks, regions, or time periods.

The geographical distribution is limited to the locations represented in the dataset.

17.2 Descriptive Nature of the Analysis

The study identifies associations between customer characteristics and churn but does not establish causal relationships.

For example, the higher churn rate among inactive customers does not prove that inactivity causes churn. Other variables may contribute to the observed difference.

17.3 Limited Product Information

The dataset records the number of banking products held by each customer but does not provide detailed information about the frequency, value, or quality of product usage.

Consequently, product count should not be treated as a complete measure of product utilization.

17.4 Small Product Groups

Only 266 customers held three products, while 60 customers held four products.

The high churn rates in these groups should be interpreted cautiously because relatively small groups may produce unstable estimates.

Further investigation using additional data would help determine whether these patterns are consistent.

17.5 Custom Relationship Strength Index

The Relationship Strength Index was created for this project using equal weights for engagement and product usage.

The index has not been independently validated and may not capture every aspect of customer relationship strength. Its results should therefore be treated as exploratory.

17.6 Limited Statistical Testing

The reported findings are primarily based on descriptive statistics and comparisons of group-level churn rates.

Formal statistical tests have not been reported in this paper. Consequently, the analysis does not establish whether the observed differences are statistically significant.

17.7 No Validated Churn Prediction Model

The project focuses on exploratory analysis, customer segmentation, and dashboard development.

It does not establish a validated machine-learning model for predicting future churn. The results describe patterns in the available dataset rather than providing verified predictions of future customer behaviour.

17.8 Potential Differences in Dashboard and Notebook Calculations

Analytical definitions and thresholds should remain consistent across the notebook and dashboard.

For example, the high-balance classification should use the same threshold wherever the high-balance group is analysed. If the threshold is recalculated after filters are applied, the displayed group may differ from the group used in the original analysis.

Consistent calculations are important for ensuring that the results can be reproduced and interpreted correctly.

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18. FUTURE SCOPE

The project can be extended in several ways.

First, additional statistical tests could be used to examine whether differences in churn rates across customer groups are statistically significant.

Second, more detailed customer segmentation could combine engagement, product utilization, account balance, tenure, and other relevant characteristics.

Third, the Relationship Strength Index could be tested using alternative weights and compared with individual customer indicators. Its usefulness would need to be evaluated against appropriate retention outcomes.

Fourth, future work could examine changes in customer engagement and churn over time if suitable longitudinal data become available.

Fifth, additional analysis could investigate salary-balance relationships and other financial characteristics to identify patterns that are not captured by the current segmentation.

Sixth, a future version of the dashboard could include additional validation checks, more detailed customer comparisons, and carefully tested indicators for monitoring retention.

Finally, a separate predictive analytics extension could be developed using suitable training and testing procedures. Such an extension would require an appropriate model, evaluation metrics, validation, and careful interpretation before it could be used to estimate churn risk.

These improvements could expand the project from descriptive analytics into a more comprehensive customer retention analytics framework.

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19. CONCLUSION

This project examined banking customer retention through customer engagement, product utilization, account balance, credit card ownership, and a custom Relationship Strength Index.

The analysis of 10,000 customer records identified an overall churn rate of 20.37%. Inactive customers had a churn rate of 26.85%, compared with 14.27% among active customers. This difference highlights the importance of examining engagement status when studying customer retention.

The product utilization analysis showed that customers holding two products had the lowest observed churn rate, while customers holding three or four products had substantially higher rates. Because the latter groups were comparatively small, their results require further investigation before broader conclusions can be drawn.

The financial commitment analysis also found that high-balance customers had a higher churn rate than lower-balance customers. Furthermore, nearly half of the high-balance group was classified as inactive. This finding suggests that account balance alone does not provide a complete picture of customer engagement.

The Relationship Strength Index provided an additional exploratory measure combining engagement and product usage. Retained customers had a higher mean index score than churned customers, although the index requires further validation before it can be used as a reliable operational measure.

The Streamlit dashboard made the results accessible through interactive filters, summary indicators, visualizations, and customer-level analysis.

Overall, the project demonstrates how Python-based analytics can be used to examine customer retention patterns and communicate findings through an interactive dashboard. Its

principal contribution is the emphasis on customer behaviour and relationship characteristics alongside financial information.

The results support a multi-factor approach to retention analysis while highlighting the need for further testing, validation, and consistent measurement. With these improvements, the analytical framework could provide a foundation for more detailed customer retention research and the development of data-informed banking strategies.

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20. REFERENCES

The following sources provide documentation for the tools and methods used in the project.

1. **Pandas Development Team.** *pandas Documentation.* Available at: https://pandas.pydata.org/docs/
2. **NumPy Developers.** *NumPy Documentation.* Available at: https://numpy.org/doc/
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7. **Unified Mentor.** *Customer Engagement & Product Utilization Analytics for Retention Strategy: Project Documentation.* Internal project documentation provided for this project.
